॥ श्रीगणेशाय नमः ॥

Programming in C++

Why C++?

- Highly portable.
- Object-oriented.
- Rich function library.
- Exception handling, function and operator overloading.
- Efficient and fast.

Features over C++

- Classes, derived classes
- Strong typing
- Inlining
- Default arguments
- Virtual functions
- Function name and operator overloading
- Polymorphism

The "Hello World!" program



Header files and functions

<iostream>:
C++ input/output streams are primarily defined by <iostream>, a header file that is part of the C++ standard library.
<iostream> uses objects cin, cout, cerr, and clog for sending data to and from the standard streams.
As part of the C++ standard library, these objects are part of the 'std' namespace.

cout is of type ostream. The cout object is said to be "connected to" the standard output device (display screen). cout is used in conjunction with the stream insertion operator << (overloads the left bit-shift operator).
cin object is of type istream uses the stream extraction operator >> (overloads the right bit-shift operator).
endl inserts a newline into the stream and calls flush.

```cpp
// example for cin, cout, endl
int main() {
    char name[100];
    std::cout << "Enter your name: ";
    std::cin >> name;
    std::cout << "Hello, " << name << "!" << std::endl;
    return 0;
}
// output:
// Enter your name: Sanved
// Hello, Sanved!
```

Namespaces

A namespace is a set of signs (names) that are used to identify and refer to objects. Namespaces are commonly structured as a hierarchy to prevent naming conflicts.

They are used to organize code into logical groups and to prevent naming conflicts when the codebase uses multiple libraries.
All identifiers within a namespace are visible only within that namespace unless explicitly qualified with the namespace name. E.g. std::cout

The using directive is used to bring the entire namespace into the current scope.
E.g. using namespace std;
On using this directive, the std:: prefix is no longer required. E.g. cout << "Hello, World!" << endl;

Variables

Variables are containers for storing data values.

Data types

In C++, there are different types of data types declared with different keywords.
int: stores integers (whole numbers), without decimals, such as 123 or -123.
float, double: stores floating-point numbers, with decimals, such as 19.99 or -19.99.
char: stores single characters, such as 'a' or 'B'.
string: stores text, such as "Hello, World!".
bool: stores values with two states: true or false.
void: represents the absence of type.